Multi-source land-use emissions reveal rising airborne fraction

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Abstract The airborne fraction is the share of anthropogenic carbon dioxide emissions that remains in the atmosphere, and a key indicator of carbon-cycle response under continued emissions. Whether this share is rising remains debated because inference is sensitive to uncertainty in land-use and land-cover change (LULC) emissions. We address this by using all available LULC measurement series constructed from Global Carbon Budget 2025 data and estimating trends with a mixed-effects model with random intercepts and random slopes by LULC series. We find that airborne fraction has increased over time, and that this conclusion is robust to excluding the final year and to benchmarking against a measurement-error-aware weighted least squares estimator. These results strengthen evidence that an increasing share of emitted carbon dioxide is accumulating in the atmosphere rather than being taken up by land and ocean sinks, with direct implications for carbon-budget assessment and near-term mitigation requirements.

Background

Whether the airborne fraction (AF) is increasing or approximately constant remains contested[1–7]. In its classical form, AF is a yearly ratio of atmospheric growth to total anthropogenic emissions, computed as the sum of fossil fuel emissions and land-use and land-cover change emissions:

$$AF_t = \frac{G_t}{FF_t + LULC_t}, \tag{1}$$

where G_t is the annual atmospheric CO_2 growth, FF_t is fossil fuel emissions excluding carbonation, and $LULC_t$ is land-use and land-cover change emissions. AF is a key carbon-cycle diagnostic, with implications for carbon-cycle feedbacks and near-term mitigation planning[1,2,8].

A persistent concern is that AF inference depends on the treatment of land-use and land-cover change (LULC) emissions, which are uncertain and model-dependent in annual carbon-budget accounting. The Global Carbon Budget (GCB) 2025[8] provides one column of LULC emissions as the average of three bookkeeping models (BLUE[9], OSCAR[10], LUCE[11]), but a broader set of model-based LULC alternatives can be constructed from the same source data. If the information from this broader set of LULC measurements is not incorporated into AF trend inference, tests can be underpowered and inference on trend direction becomes less reliable.

Here we address that issue with a design that uses all repeated LULC measurements in a mixed-effects trend framework to obtain more reliable inference. The main specification estimates AF trends from a panel of yearly AF values constructed from each LULC measurement series, with random intercepts and random slopes capturing between-series heterogeneity and inducing within-series dependence over time. As a robustness exercise, we also estimate a two-stage measurement-error weigthed least squares (WLS) model that propagates denominator uncertainty from repeated LULC measurements into annual AF variance and uses heteroskedasticity-and-autocorrelation-consistent (HAC) inference[12–14].

Across specifications that incorporate the multi-source LULC information, we find robust evidence that AF has increased over 1959-2024, and that this conclusion is not driven by the final observation (2024), which shows a large AF jump. By contrast, ordinary least squares (OLS) specifications that do not incorporate denominator measurement information provide weaker evidence of a positive trend and are more sensitive to endpoint exclusion. Taken together, the results clarify why AF trend significance has been elusive and strengthen evidence that a growing share of emitted carbon dioxide is accumulating in the atmosphere.

Data

We use annual Global Carbon Budget 2025 data for 1959-2024, with atmospheric growth G_t from NOAA/ESRL global concentration trends[15], fossil emissions excluding carbonation FF_t from the Global Carbon Project fossil dataset[8], and a panel of 69 LULC measurements per year: BLUE[9], OSCAR[10], LUCE[11], and peat-augmented process-based land-model combinations drawn from the GCB model ensemble[16–36], combined with peat components[37–39] (see Methods). The data is shown in Figure 1.

Two LULC means constructed from the panel are shown. First, the GCB LULC mean, defined as the mean of the three bookkeeping models BLUE, OSCAR, and LUCE, corresponds to the LULC column used in the Global Carbon Budget. Second, the all-LULC mean, defined as the cross-series mean from all 69 LULC measurements. The literature has typically used the GCB LULC mean as the denominator in AF trend estimation, but we show that incorporating the full set of LULC measurements materially changes inference on AF trends.

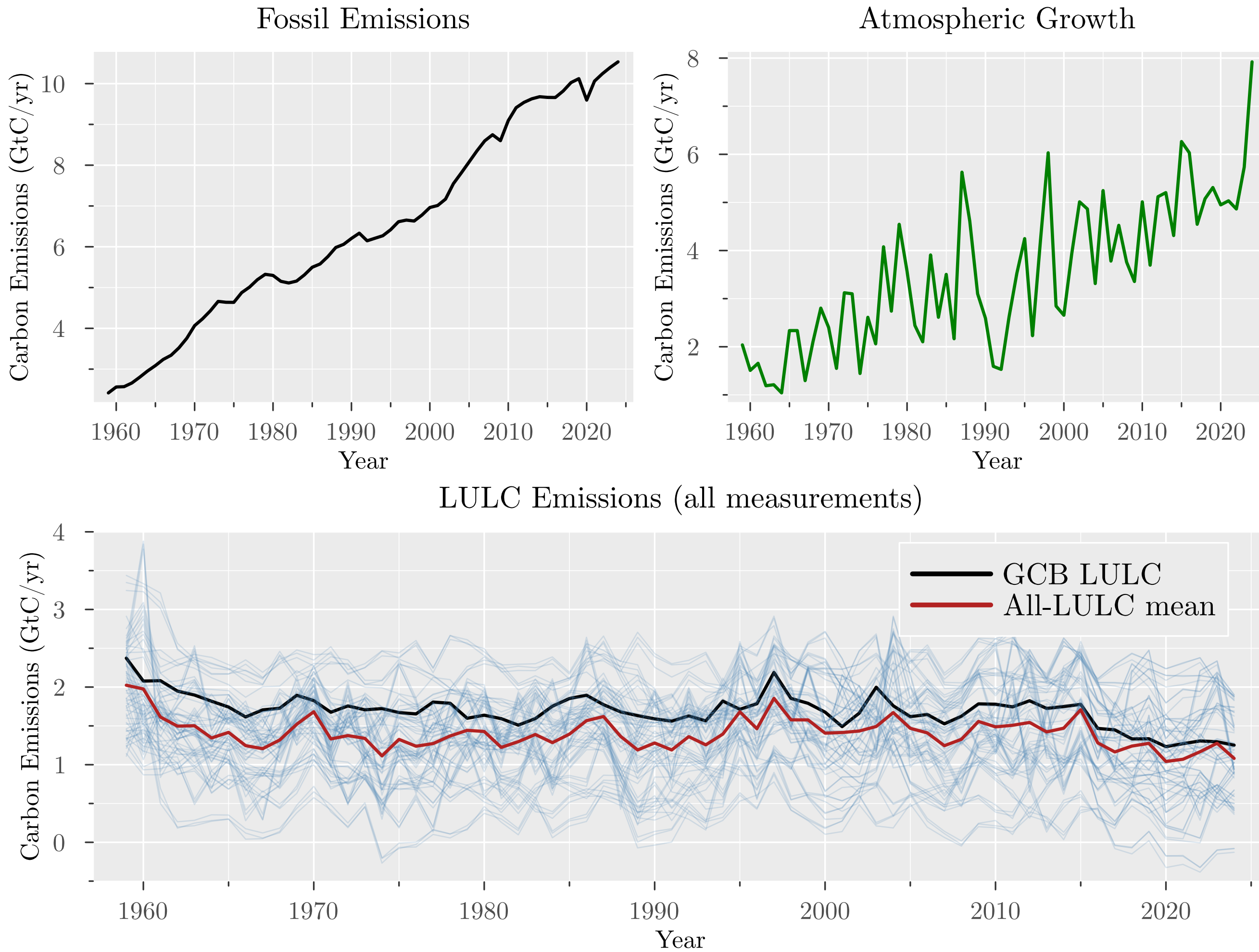


Figure 1: Global Carbon Budget 2025 data for fossil emissions, atmospheric growth, and LULC emissions (1959-2024). For the LULC panel, we show all extracted and derived LULC measurements together with the GCB LULC column and the all-LULC cross-series mean.

Identification strategy

The empirical question is whether AF has a positive linear trend. The key design choice is how to use the full panel of repeated LULC measurements while accounting for within-series dependence and cross-series heterogeneity. OLS on a collapsed annual series discards panel information and hence provides less reliable inference, see results in Table 2. Our preferred specification is a mixed-effects model with random intercepts and random slopes by LULC series, allowing correlation across years within each series and capturing unobserved heterogeneity in LULC measurements[40,41] (see Methods).

For each LULC measurement series, we construct a series of AF estimates by year (Equation 1), and the model estimates a common time trend across all series while allowing for series-specific intercepts:

$$AF_{t,j} = \alpha_j + \beta_j t + \varepsilon_{t,j},$$

where

$$\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2), \quad \beta_j \sim N(\mu_\beta, \sigma_\beta^2), \quad \varepsilon_{t,j} \sim N(0, \sigma_\varepsilon^2).$$

Here μ_α and μ_β are the fixed effects (overall intercept and slope), while σ_α^2 and σ_β^2 capture the variability in intercepts and slopes across LULC measurement series. The error term $\varepsilon_{t,j}$ captures idiosyncratic noise. The key parameter of interest is μ_β, which captures the average trend across all LULC measurement series.

Results

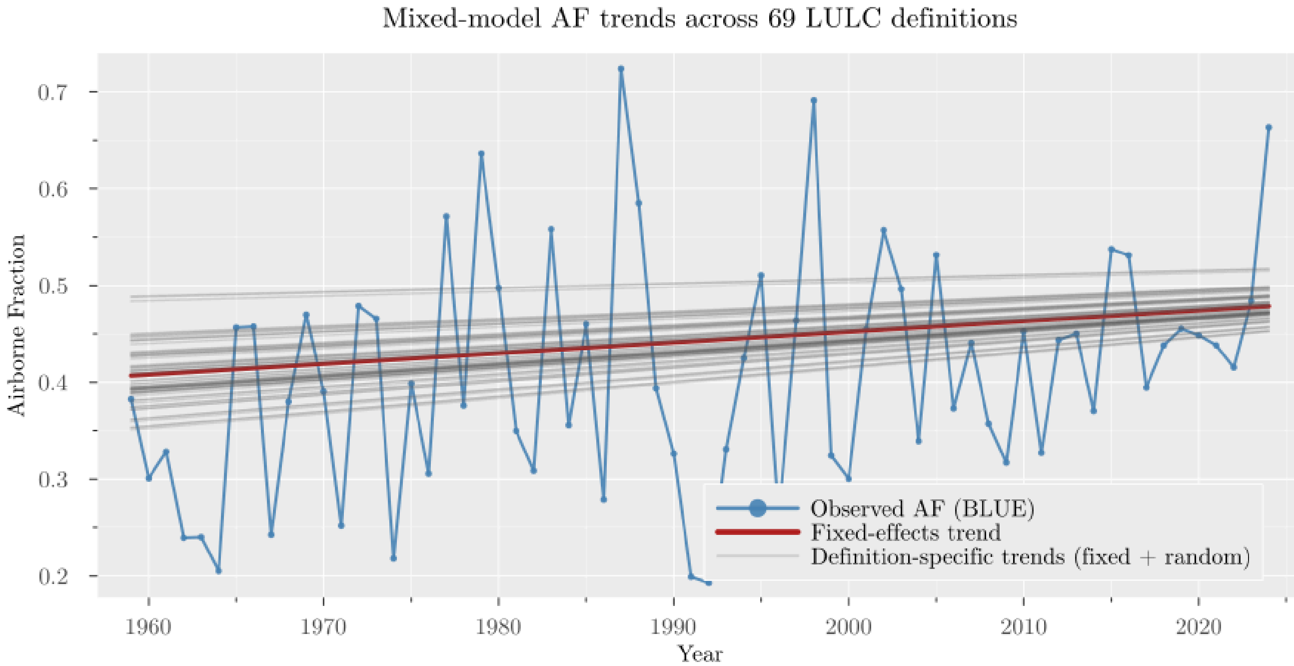
Main estimates

The main results of estimating the mixed-effects model considering the full sample (1959-2024) are shown in Table 1 and the associated trend lines are shown in Figure 2a.

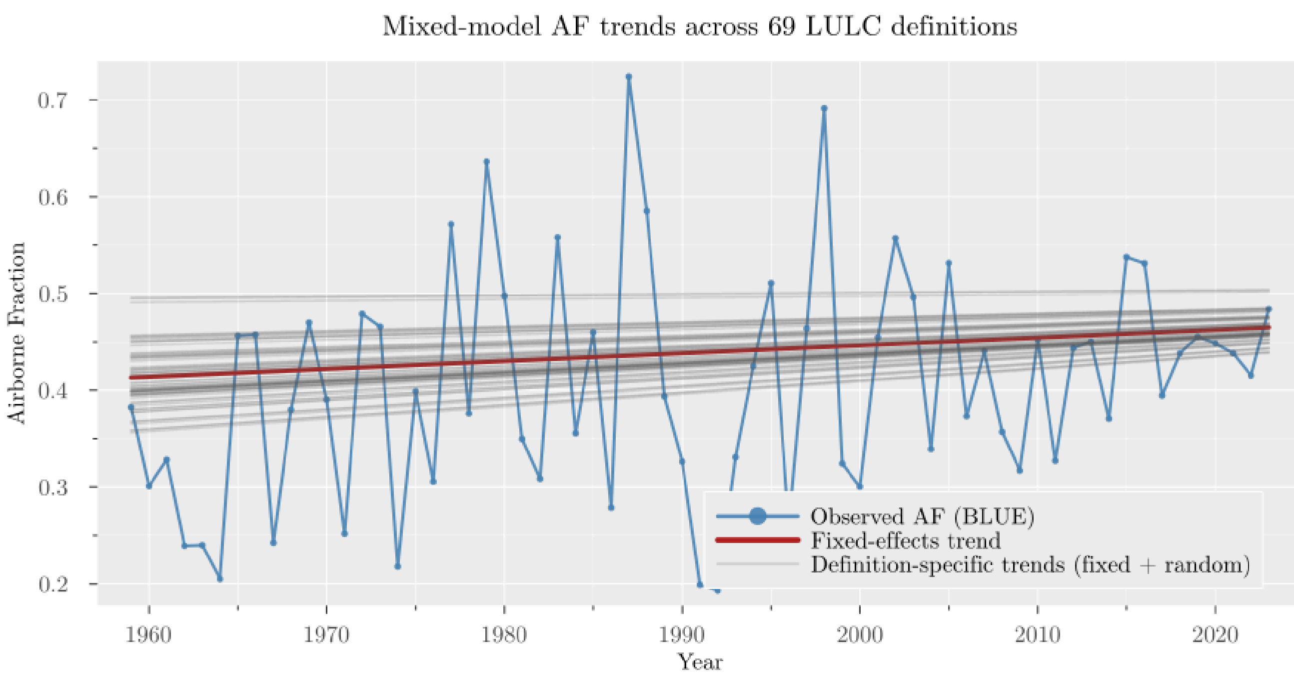
Table 1: Mixed-effects model with random effects for each LULC measurement series.

Parameter	Intercept (full)	Slope (full)	Intercept (up to 2023)	Slope (up to 2023)
Estimate	0.406959	0.001103	0.41326	0.000808
Standard error	0.005573	0.000102	0.005586	0.000103
p-value	0.00000	0.00000	0.00000	0.00000
R-squared (marginal)	0.008631	0.008631	0.004518	0.004518
R-squared (conditional)	0.700967	0.700967	0.708173	0.708173

The mixed-effects slope is positive and statistically significant (p-value < 1E-5), indicating strong evidence of an increasing AF trend across LULC measurement series. The estimated intercept is around 0.41 and the slope is around 0.0011 per year, implying an increase of about 0.07 in AF over 1959-2024. These estimates are comparable with the approximately 0.45 AF level reported in the literature[2,3,6,7,42,43], while identifying a positive long-run trend when all LULC information is incorporated. The conditional R-squared values indicate that most explained variation is captured by between-series structure rather than by the fixed linear time trend (marginal versus conditional R-squared).



(a) AF series with mixed-effects model trend (full sample, 1959-2024).



(b) AF series with mixed-effects model trend (sample ending in 2023).

Figure 2:

Figure 2a shows the AF series for each LULC measurement together with the overall mixed-effects trend line. The figure shows a clear positive trend across the panel of AF series, with the mixed-effects model capturing this common signal while allowing for series-specific heterogeneity.

Endpoint and model specification robustness

AF shows a large jump in the final year (2024), so we re-estimate the mixed-effects model on the subsample ending in 2023 to check whether the result is driven by that observation. Results are shown in Table 1 and Figure 2b. The slope remains positive and statistically significant (p-value < 1E-5), indicating that the increasing AF trend is not driven by the final observation. The slope is smaller (0.000808 per year versus 0.001103 per year), consistent with a contribution from the 2024 jump, but the direction and inference are unchanged.

As additional robustness checks, we estimate a delta-method uncertainty proxy for each annual AF estimate using the cross-measurement dispersion of the full 69-series LULC panel, and use these proxies in WLS with HAC inference[12–14,44]. Two WLS specifications are shown in Table 2: one using the GCB LULC mean as the denominator and one using the all-LULC mean. In both WLS specifications in Table 2 weights are constructed from the all-LULC panel dispersion so that the weighting scheme is held fixed across denominator definitions. The WLS results are consistent with the mixed-effects results, and Supplementary Information confirms robustness to endpoint exclusion.

Table 2: Full-sample trend comparison across OLS and WLS specifications.

Trend, full sample	OLS (GCB)	OLS (all)	WLS (GCB)	WLS (all)
Estimate	0.00158	0.00117	0.00265	0.00230
Standard error	0.00077	0.00079	0.00002	0.00002
HAC standard error	0.00062	0.00064	0.00065	0.00066
p-value	0.04021	0.13891	0.00000	0.00000
HAC p-value	0.01112	0.06846	0.00004	0.00050
R-squared	0.06171	0.03309	0.14899	0.11107

By contrast, OLS specifications that do not incorporate denominator measurement information show weaker evidence of a positive trend and greater endpoint sensitivity. In the full sample, OLS using GCB LULC is significant under both conventional and HAC inference (p-values 0.04021 and 0.01112), whereas OLS using the all-LULC mean is not significant under either method (p-values 0.13891 and 0.06846). In the sample ending in 2023 (reported in the Supplementary Information), OLS using GCB LULC is only marginally significant under conventional inference (p-value = 0.09516), and OLS using the all-LULC mean is not significant (p-value = 0.27109). This sensitivity to denominator construction and endpoint exclusion highlights the value of incorporating multi-source LULC information in AF trend estimation.

Discussion and conclusion

Using a framework that incorporates all available LULC information from Global Carbon Budget 2025, we find robust evidence that AF increased from 1959 to 2024. Methodologically, incorporating multi-source uncertainty materially changes inference relative to plain OLS. Endpoint tests show that this conclusion is not driven by the 2024 jump.

A rising AF implies that a larger share of emitted CO_2 remains in the atmosphere over policy-relevant horizons, tightening near-term mitigation requirements for a given temperature objective[1,2,8]. The increasing AF signal is consistent with broader evidence that the climate system is out of energy balance and has recently shown elevated warming and heating rates; these findings should be interpreted in that wider risk context[45–48].

Methods

Our primary estimator is a linear mixed-effects trend model with random intercepts and random slopes for each LULC measurement series[40,41,49,50]. As a robustness check, we also estimate a two-stage measurement-error trend model using repeated yearly denominator measurements, followed by WLS trend estimation with HAC inference[12–14,44,51].

Construction of the LULC measurement panel

The repeated LULC measurements are built from the Global Carbon Budget 2025 dataset. We first extract the three bookkeeping series (BLUE, OSCAR, LUCE)[9–11]. Then, for each process-based land-model LULC series in the GCB ensemble that does not already include peat emissions[16–36], we add the corresponding peat component to make it comparable to the bookkeeping models, which include peat emissions by construction. Specifically, for each of the 22 process-based series we create three peat-augmented variants by adding FAO_peat, LPX_Bern_peat, and ORCHIDEE_peat[37–39]. This produces 66 derived series, and together with BLUE/OSCAR/LUCE gives a panel of 69 yearly LULC measurements.

Mixed-effects model

The primary estimator is a linear mixed-effects trend model fitted on the panel of yearly AF values constructed from each LULC measurement series. For each year t and series j, we define

$$AF_{t,j} = \frac{G_t}{FF_t + LULC_{t,j}},$$

and estimate

$$AF_{t,j} = (\mu_\alpha + u_{\alpha j}) + (\mu_\beta + u_{\beta j})\, t + \varepsilon_{t,j},$$

with

$$\begin{bmatrix} u_{\alpha j} \\ u_{\beta j} \end{bmatrix} \sim \left(N \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma_u \right), \qquad \varepsilon_{t,j} \sim N(0, \sigma_\varepsilon^2).$$

This specification allows each LULC series to have its own level and trend while estimating a common population-average trend μ_β. The main estimand is μ_β, interpreted as the average annual change in AF across the full set of LULC definitions. Note that this notation is equivalent to the more compact notation in the main text, where $\alpha_j = \mu_\alpha + u_{\alpha j}$ and $\beta_j = \mu_\beta + u_{\beta j}$.

The model is estimated by restricted maximum likelihood to avoid bias in variance component estimation[40,41]. Fixed-effect uncertainty is based on model-based standard errors and Wald tests. We report the estimated fixed intercept and slope, standard errors, p-values, and marginal/conditional R-squared[49,50].

Identification and interpretation rely on standard mixed-model conditions: (i) conditional linearity in time, (ii) random effects centered at zero and independent across LULC definitions, and (iii) mean-zero residuals conditional on fixed and random effects.

Because shocks are common across all definitions within year, these shocks are absorbed into the time-series component of the residual term; therefore, controlled-specification checks with additional year-level covariates are treated as robustness analysis when reported.

As an endpoint robustness check, we re-estimate the same mixed-effects specification on the sample ending in 2023 to verify that inference on μ_β is not mechanically driven by the final-year observation.

Weighted least squares estimation approach

As a robustness check to the mixed-effects model, we also estimate a two-stage measurement-error-aware weighted least squares (WLS) model that uses repeated LULC measurements to construct a denominator-uncertainty proxy for each year and uses that proxy for WLS weighting in a linear trend regression with HAC inference[12–14,44,51].

In the estimation, the yearly denominator is computed as

$$\hat{C}_t = FF_t + \overline{LULC}_t, \qquad \overline{LULC}_t = \frac{1}{n_t}\sum_{j=1}^{n_t} LULC_{tj},$$

where $n_t = 3$ for the GCB LULC mean and $n_t = 69$ for the all-LULC mean. In the implemented robustness specification, the yearly weighting term is computed from the cross-measurement dispersion across the full set of 69 LULC values and is then applied to both denominator definitions.

Assume for each time $t = 1, ..., T$ we observe:

- a single numerator b_t (e.g., atmospheric CO_2 growth), and
- multiple denominator measurements $c_{t1}, ..., c_{tn_t}$ with $n_t \geq 2$, which are noisy observations of a latent C_t.

Using the repeated denominator measurements, we construct a year-specific uncertainty proxy for the ratio estimator $a_t = b_t / C_t$ via the delta method, which captures how dispersion in denominator definitions propagates into AF. The procedure is described in detail next.

We use a two-step approach to estimate the trend in the ratio $a_t = b_t / C_t$ over time, accounting for denominator measurement error.

Step 1: Estimate C_t and a_t

1. Construct the denominator estimate at time t as the sample mean across available LULC measurements:

$$\hat{C}_t = \frac{1}{n_t}\sum_{j=1}^{n_t} c_{tj}.$$

We consider two variants of this denominator estimate: one using the GCB LULC column (the mean of BLUE, OSCAR, LUCE) and one using the all-LULC mean (the mean across all 69 LULC measurements).

The cross-measurement dispersion around $\hat{C}_t$ is estimated from the multiple measurements. With $n_t \geq 2$, the empirical dispersion estimator is

$$\widehat{\mathrm{Disp}}(\hat{C}_t) = s_{c,t}^2, \qquad s_{c,t}^2 = \frac{1}{n_t - 1}\sum_{j=1}^{n_t}(c_{tj} - \bar{c}_t)^2, \qquad \bar{c}_t = \hat{C}_t.$$

2. Form the ratio estimate

$$\hat{a}_t = b_t / \hat{C}_t.$$

3. Use the multivariate delta method (shown below) to get an approximate variance of $\hat{a}_t$

$$\widehat{\mathrm{Var}}(\hat{a}_t) \propto \left(\frac{b_t}{\hat{C}_t^2}\right)^2 s_{c,t}^2.$$

Step 2: WLS regression of $\hat{a}_t$ on time

For each time t, we obtain a point estimate $\hat{a}_t$ and an uncertainty proxy that reflects cross-measurement dispersion. We use these quantities to fit a linear trend via weighted least squares (WLS):

$$\hat{a}_t = \alpha + \beta t + \varepsilon_t, \quad \varepsilon_t \sim (0, \sigma_t^2), \quad \sigma_t^2 = \widehat{\mathrm{Var}}(\hat{a}_t).$$

In our application, the time index t is the year, and $\hat{a}_t$ is the estimated AF for that year. The weighting term σ_t^2 is proportional to estimated AF uncertainty from denominator dispersion and varies across years with the spread of the LULC measurement panel. We use this term to weight the regression, giving more weight to years with lower denominator-related uncertainty.

Assuming negligible correlation in ε_t across time, this is WLS with weights $w_t = \frac{1}{\sigma_t^2}$. If we suspect serial correlation, we can use HAC standard errors for inference on β without changing the point estimate. In practice, we use a HAC covariance estimator with a Bartlett kernel and Andrews automatic bandwidth selection[14]. In the results, we report both conventional and HAC standard errors.

Testing for a time trend is then a test of $\beta = 0$ versus $\beta \neq 0$ in this linear model, and WLS uses the repeated denominator information through this delta-method weighting scheme.

Delta method for ratio variance estimation

Step-by-step derivation (first-order delta method):

Define the random vector and the function of interest as

$$X_t = \begin{bmatrix} b_t \\ \hat{C}_t \end{bmatrix}, \qquad g(X_t) = \frac{b_t}{\hat{C}_t}.$$

and let

$$\sigma_{b,t}^2 = \mathrm{Var}\ (b_t), \qquad \sigma_{C,t}^2 = \mathrm{Var}\ (\hat{C}_t), \qquad \sigma_{bC,t} = \mathrm{Cov}\ (b_t, \hat{C}_t).$$

1. Linearize g around the mean vector $(\mu_{b,t}, \mu_{C,t}) = \mathbb{E}[(b_t, \hat{C}_t)]$:

$$g(X_t) \approx g(\mu_{b,t}, \mu_{C,t}) + \nabla g(\mu_{b,t}, \mu_{C,t})^\top (X_t - \mathbb{E}[X_t]).$$

2. Compute the gradient:

$$\nabla g(b, C) = \begin{bmatrix} \partial g/\partial b \\ \partial g/\partial C \end{bmatrix} = \begin{bmatrix} 1/C \\ -b/C^2 \end{bmatrix}.$$

3. Write the covariance matrix of $(b_t, \hat{C}_t)$:

$$\Sigma_t = \begin{bmatrix} \sigma^2_{b,t} & \sigma_{bC,t} \\ \sigma_{bC,t} & \sigma^2_{C,t} \end{bmatrix}.$$

4. Apply the delta-method variance formula

$$\mathrm{Var}\ (g(X_t)) \approx \nabla g(\mu_{b,t}, \mu_{C,t})^\top \Sigma_t \nabla g(\mu_{b,t}, \mu_{C,t}),$$

   and with plug-in evaluation at (b_t, C_t), this becomes

$$\mathrm{Var}\ (\hat{a}_t) \approx \left(\frac{1}{C_t}\right)^2 \sigma^2_{b,t} + \left(\frac{b_t}{C_t^2}\right)^2 \sigma^2_{C,t} - 2\frac{b_t}{C_t^3}\sigma_{bC,t}.$$

In practice, we use plug-in estimates (replace unknown moments by their empirical counterparts). This is the standard delta-method approximation for the variance of a ratio estimator.

5. In the case analysed in this paper, we have a single numerator measurement from a robust source per time, so we treat $\sigma^2_{b,t}$ and $\sigma_{bC,t}$ as negligible relative to the variance from the denominator measurement error, which is captured by $\sigma^2_{C,t}$.

The expression simplifies to

$$\mathrm{Var}\ (\hat{a}_t) \approx \left(\frac{b_t}{C_t^2}\right)^2 \sigma^2_{C,t}.$$

Accordingly, a practical plug-in weighting proxy is

$$\tilde{\sigma}^2_{a,t} = \left(\frac{b_t}{\hat{C}_t^2}\right)^2 s^2_{c,t}.$$

Data availability

The study uses publicly available Global Carbon Budget 2025 data and derived yearly series generated from those source files. The processed analysis tables are provided in the manuscript repository under the results directory, and the extracted and derived LULC panel is available as a CSV file in the data directory.

Code availability

All code used to process data, estimate models, and generate figures is available in the repository (github.com/everval/Airborne-Fraction-WLS-Trend) under the scripts directory, including Quarto analysis files and Julia helper functions.

Competing interests

The author declares no competing interests.

Additional information

Supplementary information is available in the accompanying supplementary manuscript file.

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